PDF OPERATION MANUAL

VISA

VCA Multi-Agent Synthetic Researcher

How a Visa/VCA reviewer can upload a real survey/interview PDF as the input artifact, run Swiss synthetic customer agents with IBM watsonx.ai / Granite, and interpret segment perspectives, aggregate insights, persona-level responses, validation checks and a consultant decision brief.

What this proves

- Flexible survey/interview input: no fixed question set.
- PDF upload and extraction audit for realistic partner testing.
- Swiss public-data-grounded synthetic customer simulation.
- Customer board, aggregate insights and individual response traceability.
- Benchmark, consistency, coverage and realism validation.
- Real-model proof path using IBM watsonx.ai / Granite 4 H Small, configured locally and in Code Engine.

Review artifacts

Application:
https://visa-synthetic-research-copilot.27cqtktlikeo.eu-de.codeengine.appdomain.cloud

Repository:
https://github.com/ZihaoLu001/visa-synthetic-research-copilot

Sample PDF:
Federal Reserve mobile-payments survey excerpt adapted for public-data review use.

Live model:
IBM watsonx.ai in eu-de, default model ibm/granite-4-h-small. Code Engine health reports watsonx_configured=true.

Public-data prototype. No Visa internal or client-sensitive data is used.

START HERE

Reviewer Workflow At A Glance

VISA

Who should use this?

Visa Consulting & Analytics reviewers, IBM mentors, or student evaluators who want to test the prototype with a survey/interview artifact instead of a hardcoded prompt.

12
Default live Granite proof respondents

96
Full synthetic panel available

PDF
Recommended partner-review upload format

C
Latest watsonx example evidence grade

POSITIONING

This is an early-stage multi-agent synthetic research workbench. It helps consultants identify promising propositions, weak assumptions, message risks and real-research priorities before commissioning real customer research.

Four-step operation

1 UPLOAD
Attach a PDF survey/interview guide or use the sample public survey.

2 REVIEW
Check extracted scenario text and edit questions if needed.

3 CONFIGURE
Set target market and define the client value proposition.

4 RUN
Generate synthetic customer lens, insights, persona responses and validation evidence.

RECOMMENDED REVIEWER SETTING

Use **Model provider = watsonx** for the final real-model proof. Keep **mock** only as a fallback for rehearsal or quota issues. Use **Quick real-model proof** for a fast live run, then move to **Full survey** and **96 respondents** when quota/time allows.

DELIVERY ARCHITECTURE

Model, Algorithm And Delivery Stack

VISA

Real LLM path

IBM watsonx.ai is the primary provider. The app calls Granite through the ibm-watsonx-ai ModelInference abstraction, defaulting to ibm/granite-4-h-small in eu-de.

Fallback path

MockLLM is deterministic and exists only for CI, rehearsal and classroom quota contingency. It is not presented as real customer intelligence.

Core algorithm

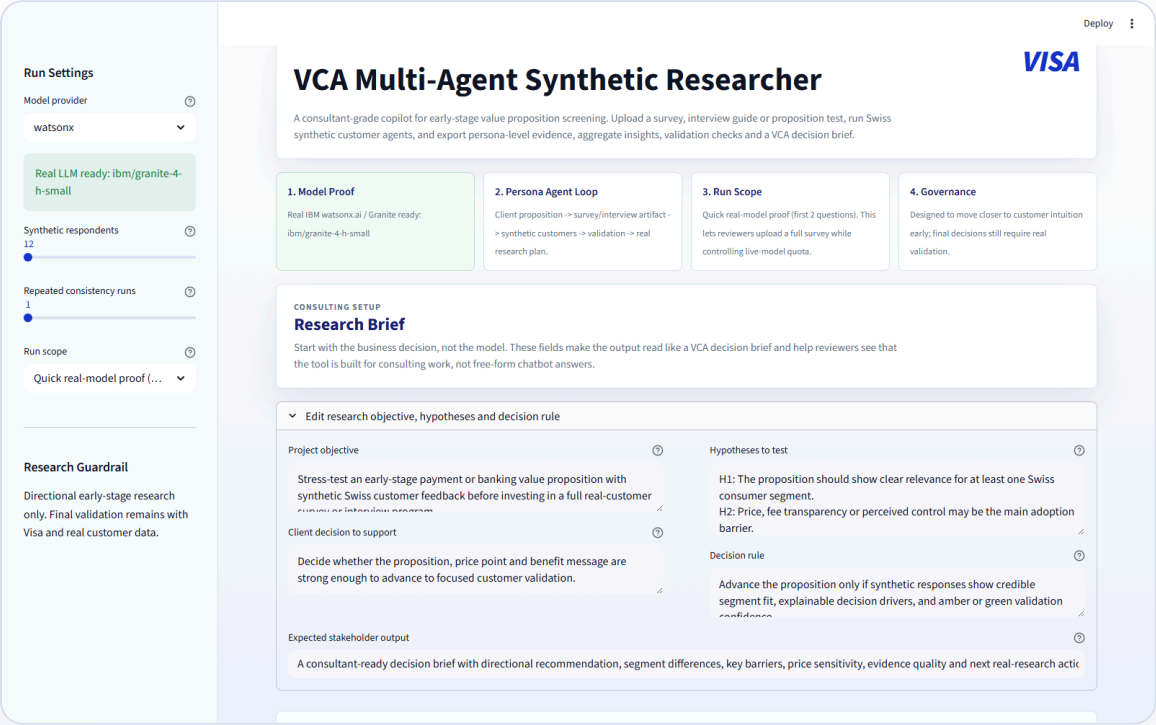
Extract survey/interview text, parse structured questions, normalize key constructs, sample Swiss micro-personas, run one synthetic customer agent per persona, aggregate weighted results, validate, and synthesize a VCA decision brief.

Customer perspective board

The survey is the input artifact; the product output is a customer board with need states, objections, message tests, decision drivers, time/cost advantage and real-customer bridge.

Consultant quality layer

The app adds evidence grade, decision risk, lead-margin interpretation, risk flags, survey repair plan and real-customer validation plan so the output is actionable rather than decorative.



Consultant deliverable

Each run exports a polished PDF report and a delivery pack with a decision brief, customer perspective JSON, quality layer JSON, consultant report, persona CSV, validation JSON, full run JSON, source audit and governance notes.

Consulting guardrail

Synthetic results are directional. They help screen propositions and design better real customer research; they do not replace final Visa validation.

INPUT ARTIFACT

Step 1 - Upload A Real Survey PDF

Preferred format

Use PDF for partner review because it matches how survey decks and interview guides are usually shared.

Sample file

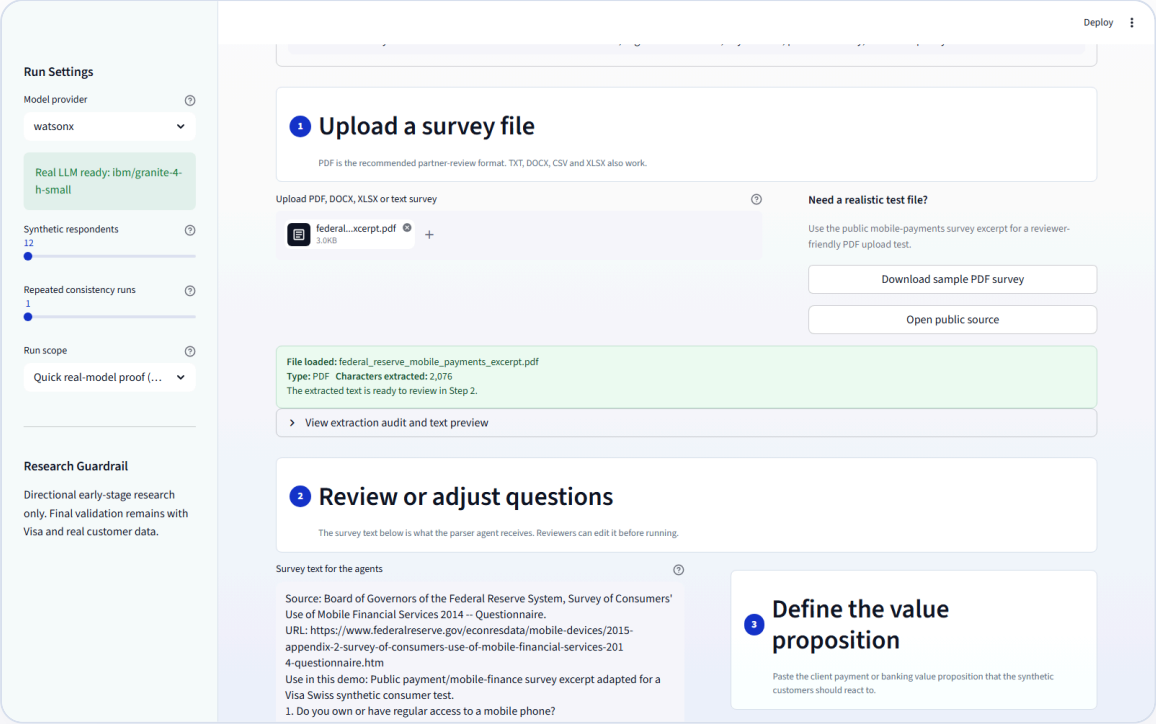
The right panel lets reviewers download the included public PDF survey excerpt and upload it immediately.

Extraction proof

After upload, the app shows file name, file type and extracted character count before any agent run.

Data guardrail

The sample comes from a public Federal Reserve questionnaire; no Visa internal data is required.



Public source: https://www.federalreserve.gov/econresdata/mobile-devices/2015-appendix-2-survey-of-consumers-use-of-mobile-financial-services-2014-questionnaire.htm

SURVEY PARSER INPUT

Step 2 - Review Or Adjust Questions

Editable extracted text

The textarea is the exact survey text sent to the parser agent. Reviewers can leave it unchanged or edit wording.

Why this matters

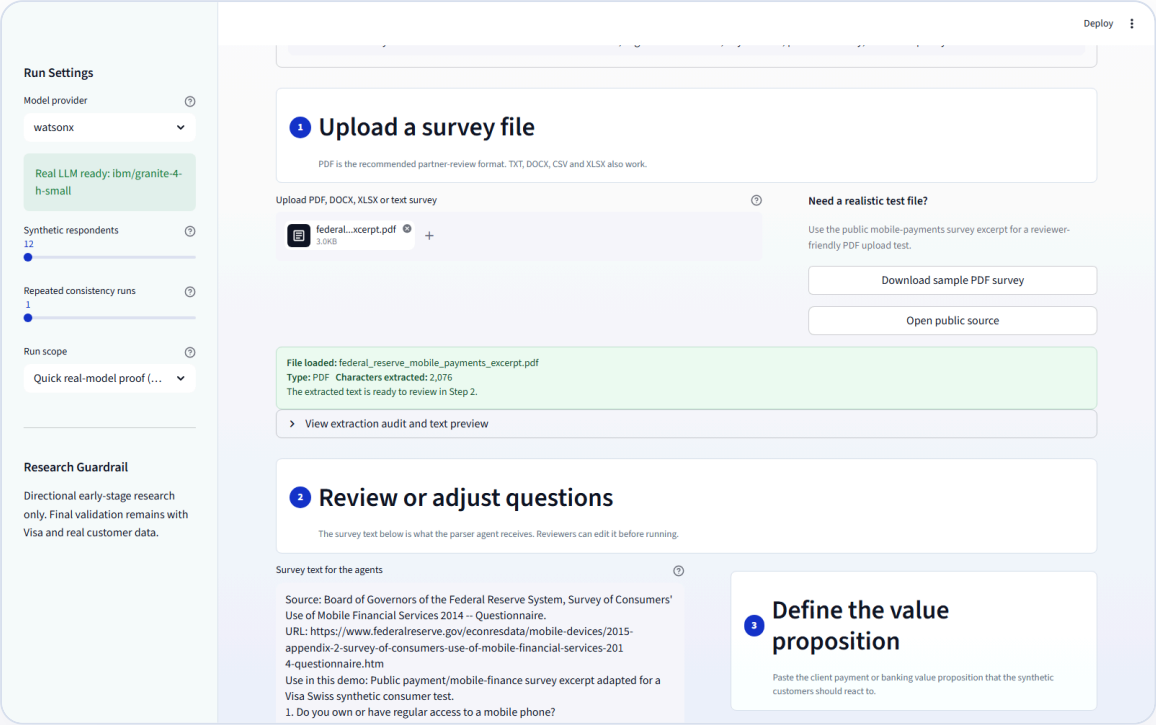
This directly addresses the Visa clarification that the system should answer survey questions without a fixed limitation.

Supported artifacts

The same ingestion layer accepts PDF, DOCX, XLSX, CSV, TXT and Markdown.

Audit trail

If the extracted text is edited, the run records that fact in the Question Parser tab.



AGENT WORKFLOW

Step 3 - Define Proposition And Run

Proposition testing

Paste the client value proposition that Visa/VCA wants synthetic customers to react to.

Live sensitivity

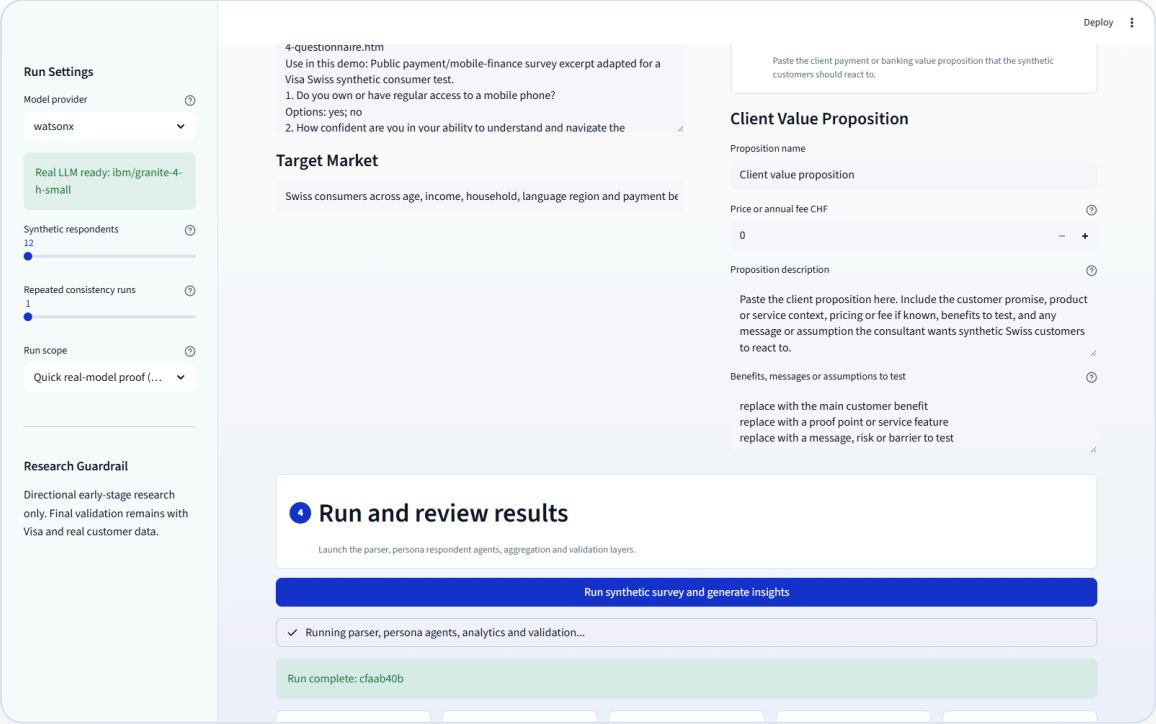
Change the price, benefit or trust message, rerun, and compare movement in segment fit and adoption index.

Panel scale

The quick watsonx proof starts small to conserve quota. The full 96-persona run expands seven Swiss archetypes into a weighted micro-population.

Run button

The button launches survey parsing, persona respondent agents, analytics, validation and decision-brief synthesis in one workflow.



AGGREGATED OUTPUT

Step 4 - Read Consultant-Level Insights

Recommendation

The first tab summarizes the proposition signal and the reason behind the result.

Adoption index

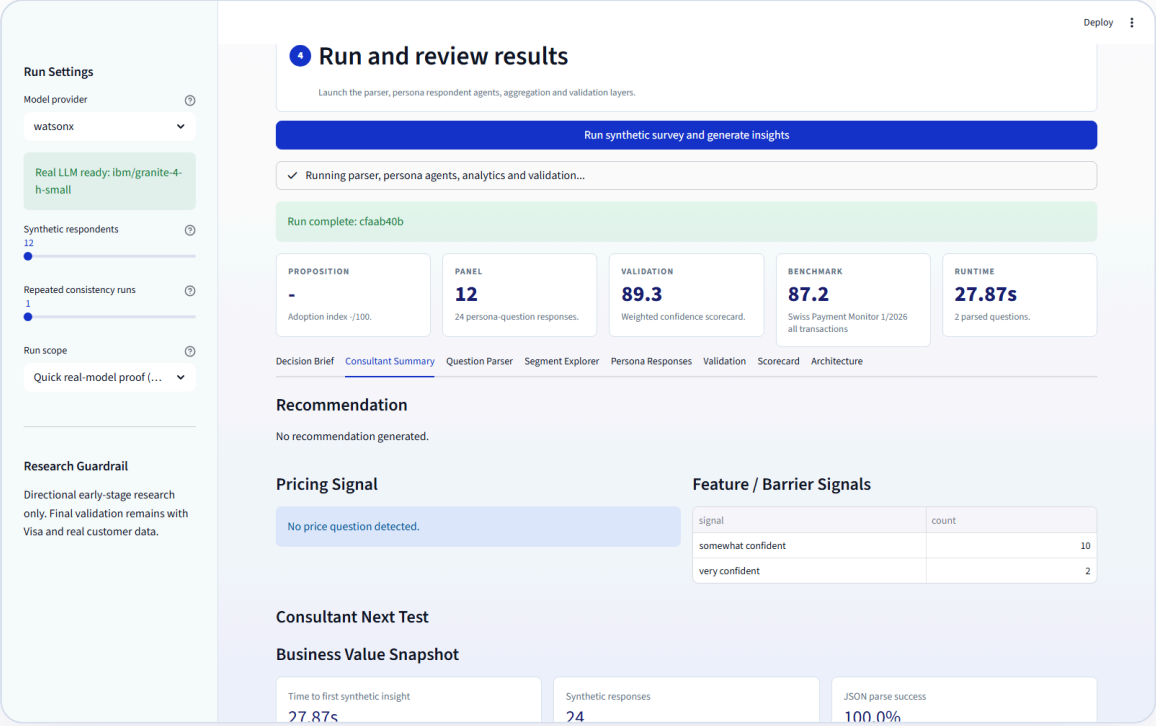
The bar chart shows proposition fit across the synthetic Swiss panel.

Pricing signal

Acceptable fee ranges help frame a real customer price-sensitivity test.

Feature and barrier signals

Top labels surface recurring needs, trust drivers and objections.



QUESTION PARSER TAB

Confirm The Survey Was Parsed From PDF

Input Source Audit

Shows Source = uploaded_file, File type = pdf, character count and extraction notes.

No hardcoding

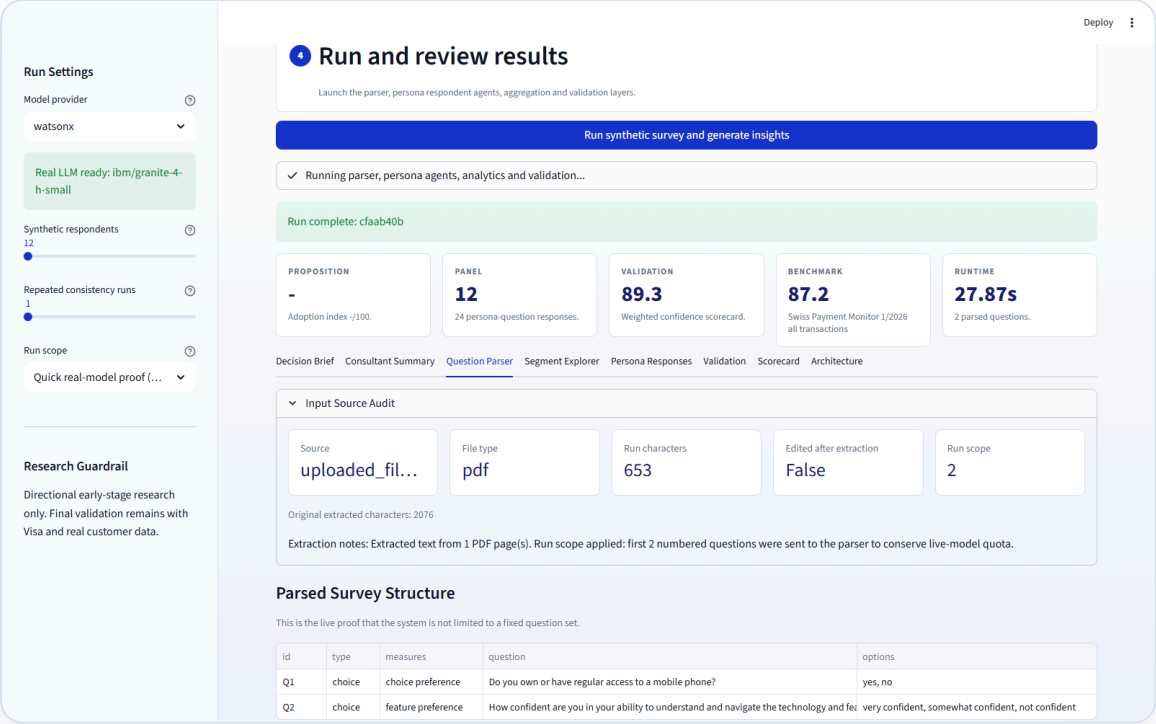
The parsed table lists question id, type, construct and options generated from the uploaded survey.

Review expectation

Open this tab during review to prove the app can accept a new survey artifact.

Quality fix

The parser handles PDF line wrapping so wrapped questions stay together.



SEGMENT EXPLORER TAB

Inspect Segment Differences

Why segments matter

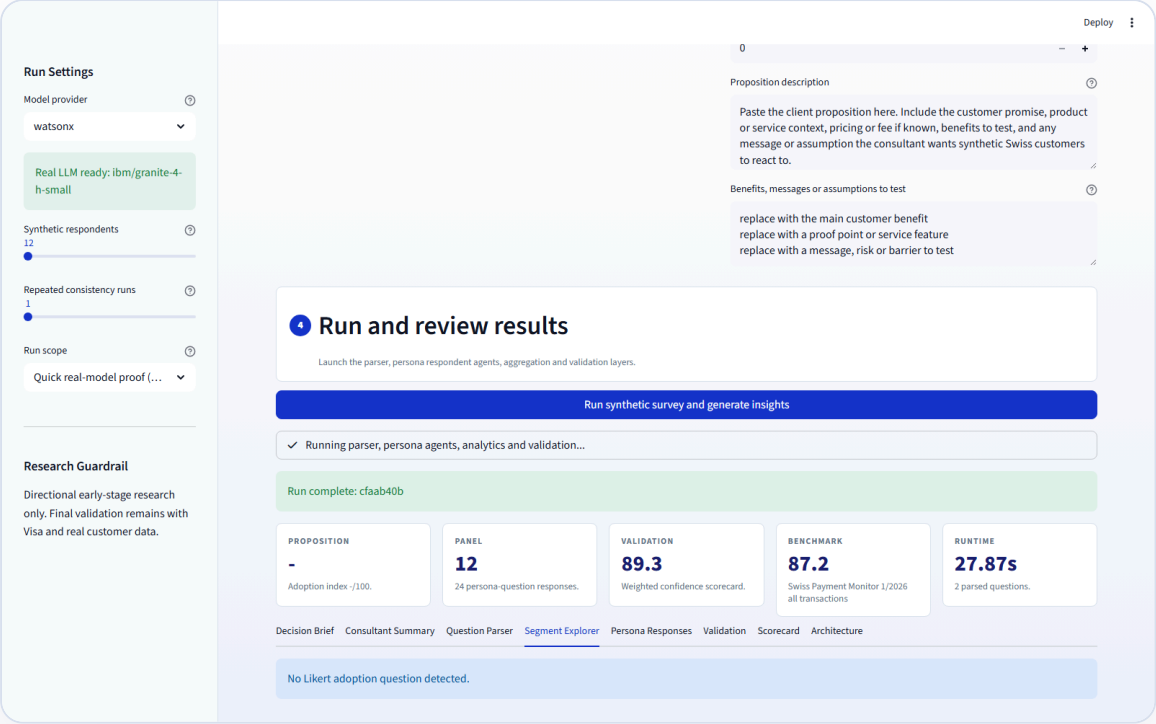
Visa asked for different personas or demographic groups, not one generic customer.

What to look for

Compare how students, family budget parents, travelers, retirees and SME owners respond differently.

Consultant use

Segment variation helps decide which customer group deserves real follow-up research.



PERSONA RESPONSES TAB

Inspect Persona-Level Traceability

Individual responses

Every row stores persona id, persona name, proposition, question, answer, rationale and confidence.

Filter workflow

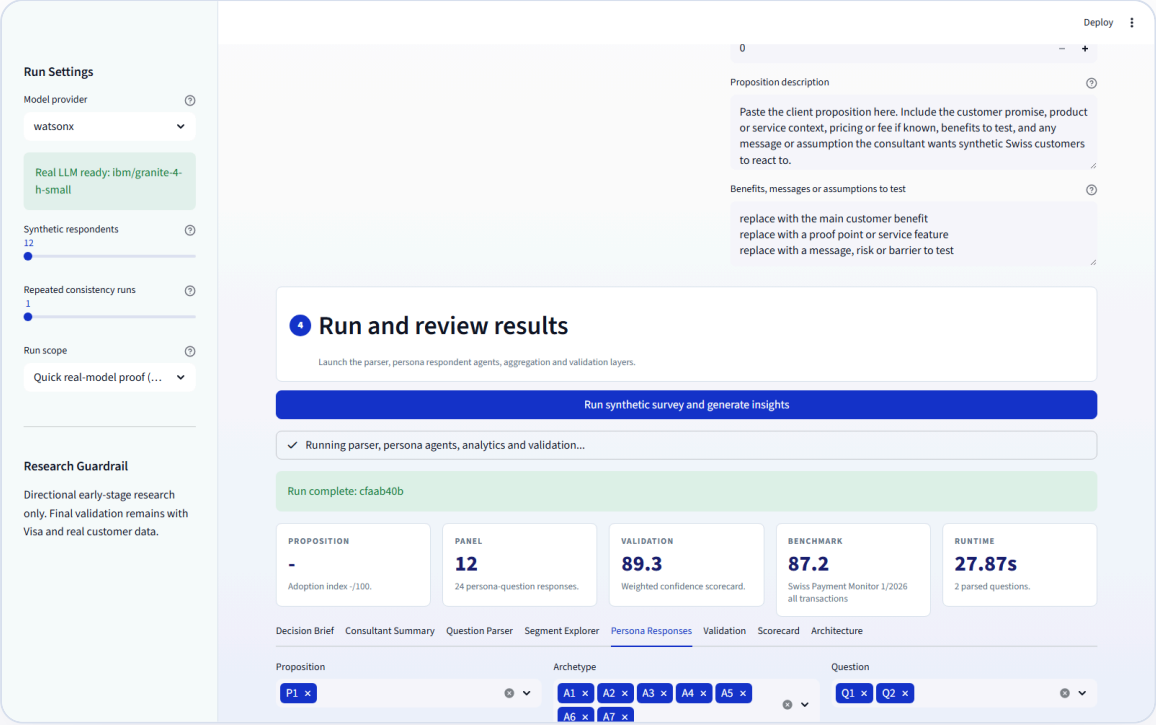
Filter by proposition, archetype or question to inspect why a segment is driving the aggregate result.

Reviewer value

This satisfies the Visa request for both aggregated insights and persona-level survey responses.

Export

Download the filtered response table as CSV for analysis or appendix material.



VALIDATION TAB

Check Validation Confidence

Benchmark alignment

Compares synthetic payment behavior mix against public Swiss payment survey anchors.

Internal consistency

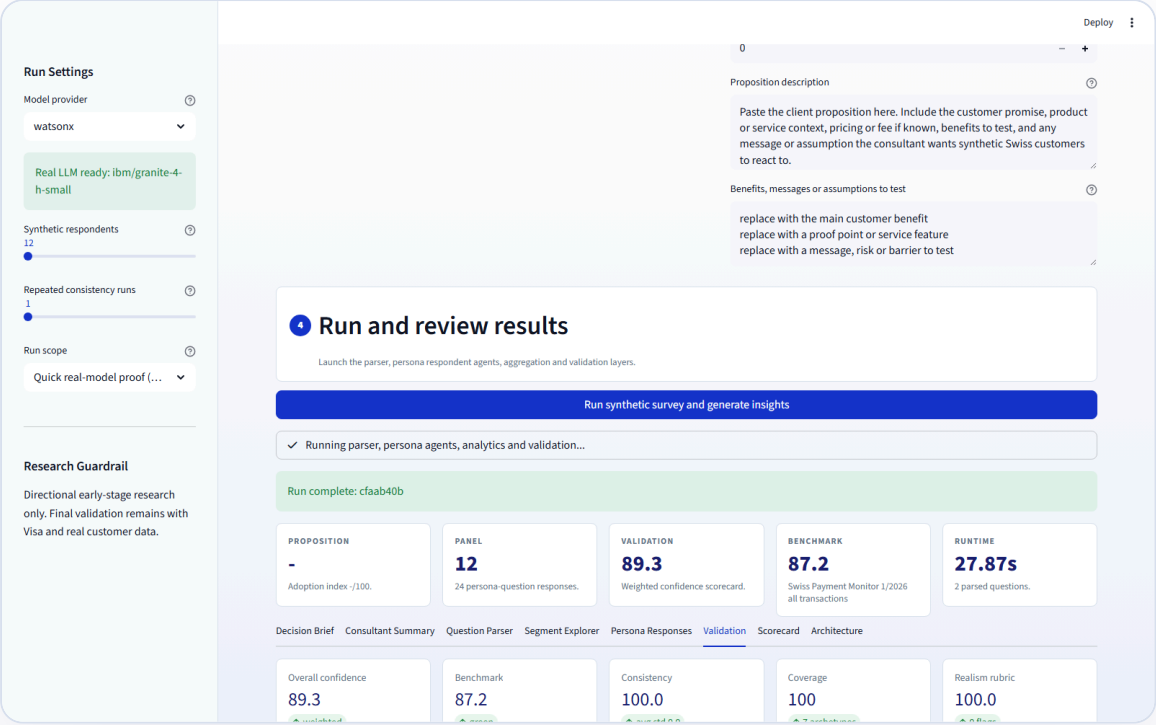
Checks repeated persona runs for response variance.

Coverage

Confirms the panel covers the Swiss archetypes and the survey covers key constructs.

Interpretation

This is directional evidence, not a claim that synthetic respondents replace real Visa validation.



DECISION BRIEF TAB

Read The Customer Perspective Board

Why this matters

This is the core Visa output: synthetic customer perspectives for early-stage value proposition decisions.

Customer board

Summarizes each Swiss segment's reality, proposition fit, need state, objections to probe and message to test.

Decision drivers

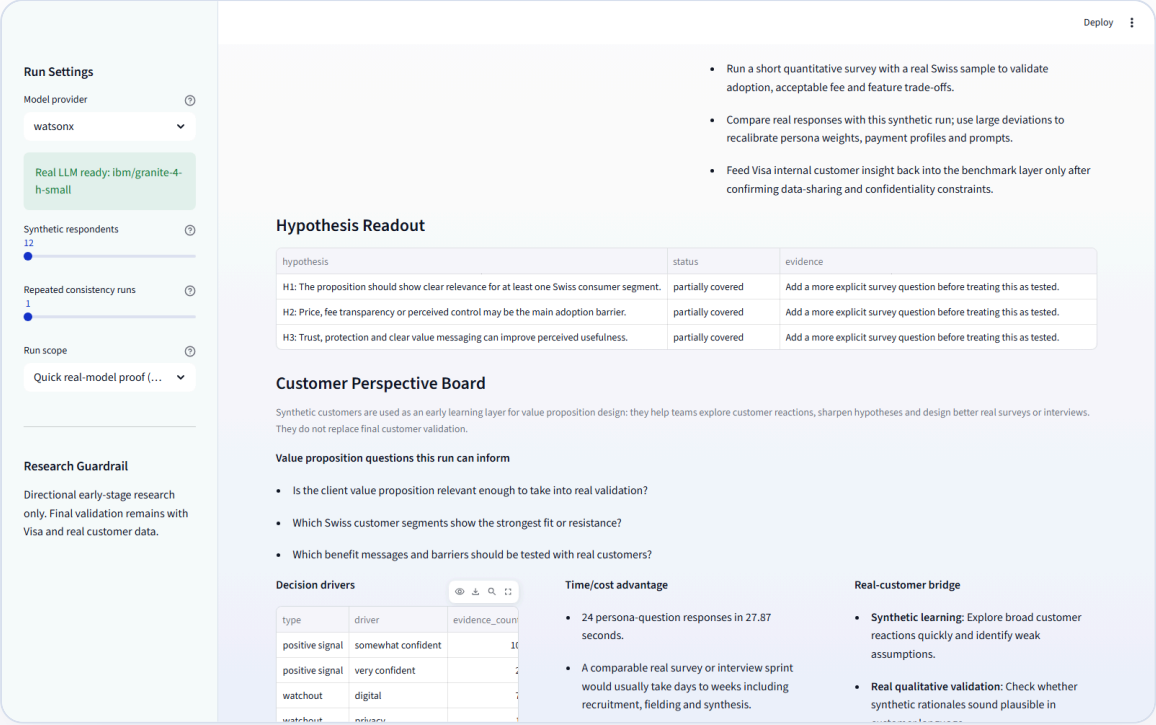
Shows the positive signals and watchouts that explain the aggregate recommendation.

Time and cost advantage

Shows why the synthetic layer is useful before commissioning real surveys or interviews.

Real-customer bridge

Keeps the governance clear: synthetic learning first, qualitative validation next, quantitative calibration after that.



DECISION BRIEF TAB

Read The Consultant Quality Layer

Evidence grade

Summarizes whether the synthetic run is strong enough for directional prioritization or still needs major repair.

Decision risk

Combines validation, lead margin and segment differentiation so consultants do not over-read a narrow synthetic lead.

Risk flags

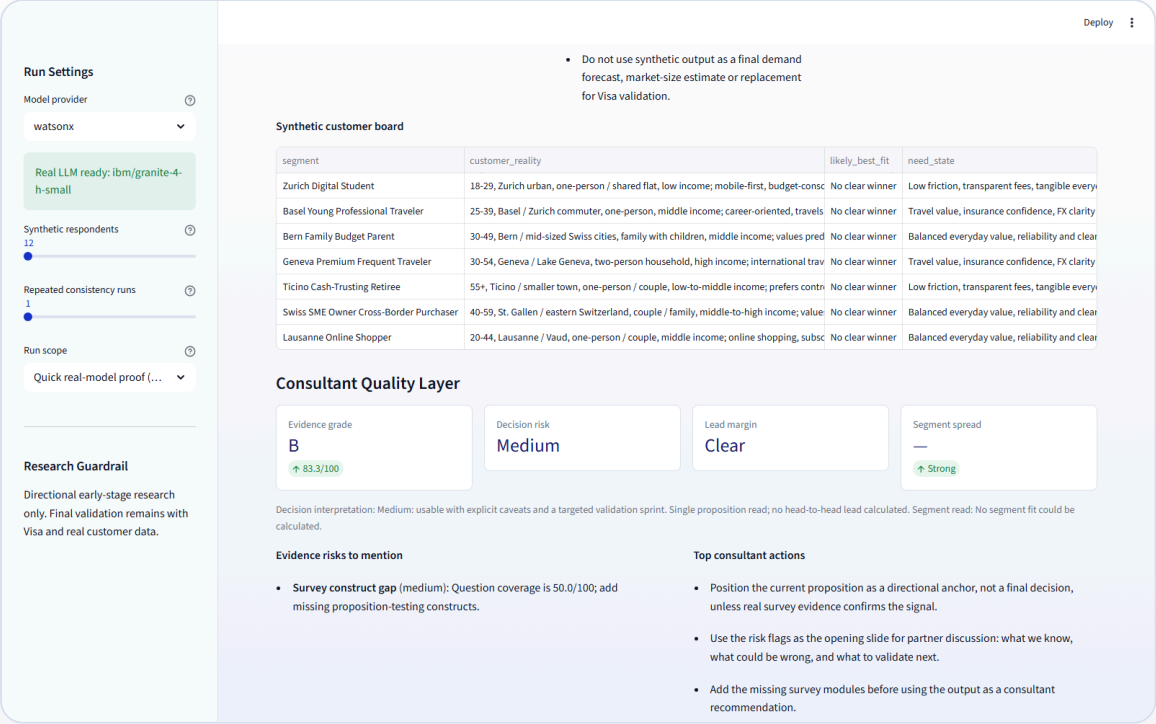
Calls out weak lead, construct gaps, realism issues or price-value mismatch in plain business language.

Survey repair plan

Recommends exact modules to add before a real Swiss customer survey, such as choice-based validation, price sensitivity, feature trade-off and barrier/trust questions.

Real validation plan

Turns the output into the next consulting action: interviews, Swiss pulse survey, pricing cell test and calibration loop.



EVALUATION MAPPING

Use The Scorecard In The Final Presentation

Pilot readiness gate

The top of the Scorecard tab shows which checks are ready for partner review and which need attention.

Real model status

For final proof, Evidence mode should show Real IBM watsonx.ai and ibm/granite-4-h-small.

Live product proof

Running app, PDF upload, parser, agents, results and validation are all live.

Architecture

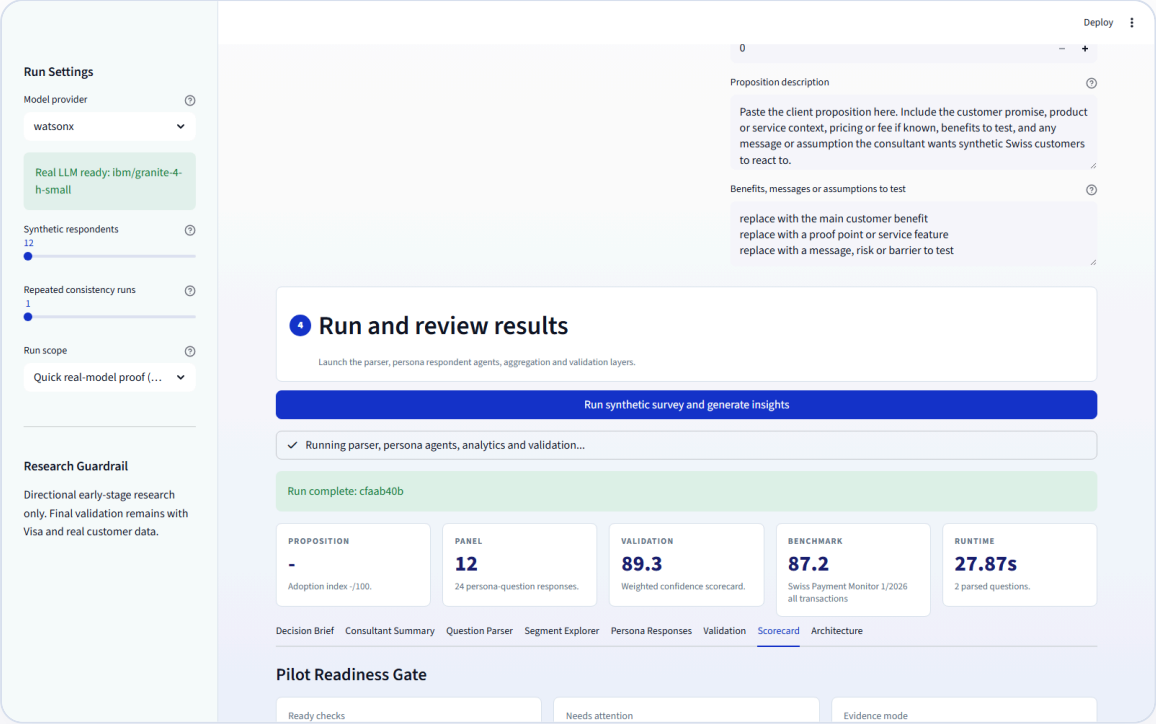
UI, ingestion, parser, persona store, orchestrator, respondent agents, analytics and validator are visible.

KPIs

Runtime, response count, parse success, consistency and benchmark scores are quantified.

Business value

Position the system as early-stage proposition screening and better real-survey design.



PORTABLE PARTNER ARTIFACT

Export PDF Report And Delivery Pack

Why this matters

VCA reviewers should not need to trust only the live screen. They can download a consultant-style PDF report and inspect the evidence offline.

PDF report

The report includes executive answer, Customer Perspective Board, consultant quality layer, proposition decision matrix, segment fit, persona evidence, validation confidence, methodology and limitations.

Included files

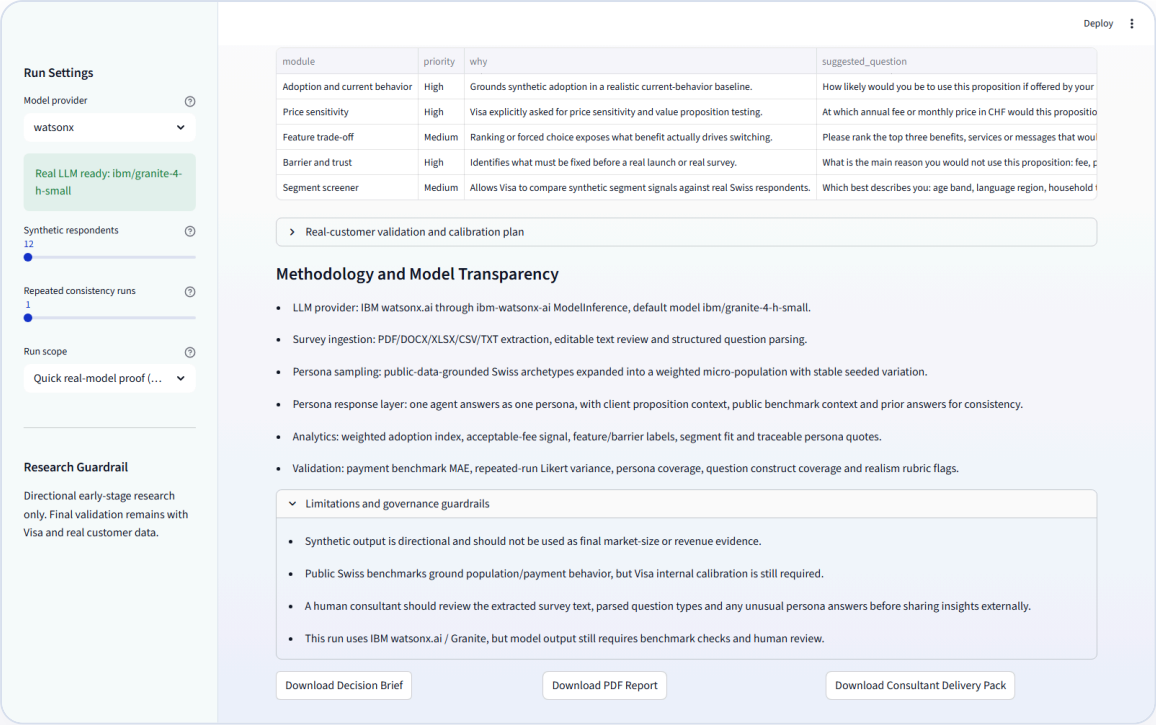
The ZIP contains the PDF report, decision brief, customer perspective JSON, quality layer JSON, consultant report, persona-level CSV, validation JSON, full run JSON, input-source audit, methodology/governance notes and pilot-readiness gate.

Review workflow

Send the input PDF survey plus output PDF report together with the app URL so IBM/Visa can check recommendation logic, persona traceability and validation evidence.

No secrets

The delivery pack contains run evidence, not watsonx API keys or local credentials.



PARTNER ALIGNMENT

Suggested Slack Message To Visa / IBM

Message draft

Hi Visa / IBM team,

We prepared a short operation manual and a sample input/output pair for the current prototype. It shows a real PDF survey/interview artifact upload, dynamic parsing, IBM watsonx/Granite model path, Swiss synthetic customer responses, a segment customer board, aggregated consultant insights, a Consultant Quality Layer, validation checks, and a generated consultant PDF report.

Attached are the input PDF survey and the generated output PDF report from the same flow.

Could you please let us know whether this matches your expected direction for the final Visa use case?

1. Is the Multi-Agent Synthetic Researcher framing aligned with your intended use case?
2. Is the flexible PDF survey/interview upload flow aligned with your expectation as the input artifact?
3. Should the final output emphasize aggregate insights, persona-level responses, synthetic customer segment cards, or all three?
4. Is the evidence grade / decision risk / survey repair layer useful for VCA consultants?
5. Are the benchmark, consistency, coverage and realism validation checks useful enough for VCA consultants?
6. Is there any missing Visa requirement we should address before finals?

Thank you!

What to emphasize verbally

- **Flexible input:** The system is not a fixed Q&A bot.
- **Public-data grounding:** Personas are used because internal Visa customer data is unavailable.
- **Consultant value:** It produces early directional evidence, a decision posture, evidence quality and better real survey design.
- **Validation discipline:** The app exposes limitations through benchmark and consistency checks.

BEST CHANNEL

Use the Visa-specific Slack channel for use-case/product feedback. Use the general guest channel for deployment or IBM platform issues.